AR SHEET METAL DESIGN BEST PRACTICES

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Hardware Engineering

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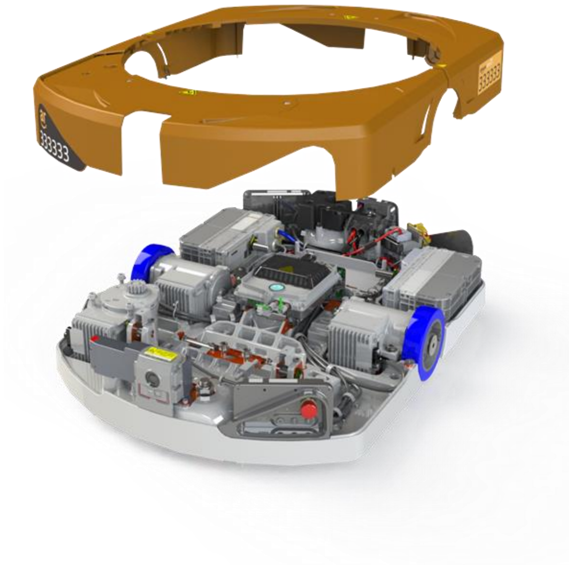

Amazon Robotics has a high standard for engineering design quality. Mistakes made during design process become very costly if they escape to the field. This document contains a description of most sheet metal fabrication processes, mechanical design practices (DFM) that will shorten the design process, reduce design defects, and promote peer review & collaboration.

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CHAPTER 1: SHEET METAL INTRODUCTION

Introduction – Sheet Metal Advantages

- High level of dimensional stability
- Repeatable Process
- High range of materials and properties available
- Prototypes are representative of parts produced with production processes
- Flexibility for change if areas are known early in design/tooling process
- Provides RFI and EMI shielding
- Thermally conductive

Introduction – Sheet Metal Disadvantages

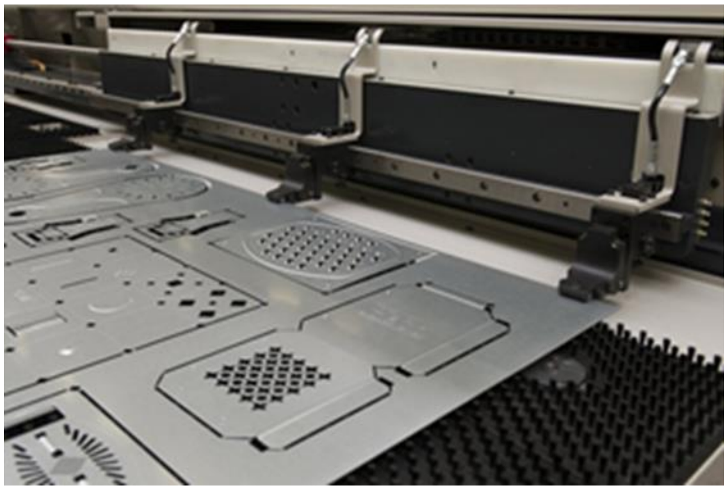
- Higher weight than some technologies
- Feature geometry is limited
- Thermal growth
- Acoustics attenuation

CHAPTER 2: FABRICATION PROCESS AND CAPABILITIES

Sheet Metal Fabrication Processes

- Soft Tooling (Early prototype and Alpha Builds)
  - Turret Press
  - Laser cutting
  - Hand press brake forming
- Hard (Tooling Beta -> GA)
  - Punch Press
  - Progressive tooling
  - Multi Slide
  - Fine Blanking

Turret Press



Turret Press Fabrication Process

A turret press, is a CNC metal fabrication tool used to process flat blank sheet metal using punches to form shapes. Turret punching involves pushing a forming die or a punch into or through a piece of metal to create a form or a through hole. The turret punch can make forms and holes in a variety of different profiles and diameters and can make as many features in a sheet as long as the structure of the sheet is maintained. The material is moved relative to a fixed rotary upper and lower punching head that can contain as many as 60+ unique punch profiles. Turret punching is an extremely fast and affordable way to make holes in sheet metal. Due to the flexibility of the process it is well suited for fabrication of prototypes and low volume production runs. The turret punch is usually very cost-effective because it contains multiple tooling in one device, meaning manufacturers avoid the expense of creating multiple, job-specific punch tools sets.

Turret Press Process Pros/Cons

Pros

- Highly flexible process for both holes and shallow forms (counterbores, etc.)
- Easy to setup
- Requires limited operator interaction during operation
- Inexpensive tooling for custom hole profiles of shapes
- High punch rate between 80-300 hits/minute
- Larger features can be made with multiple hits

Cons

- Can only fabricate close to flat forms
- Limited material thickness capability
- Higher parts cost compared to progressive process

Laser Cutting

Laser Cutting Process

Laser cutting is an accurate and fast fabrication process using high-powered laser to make precision cuts in sheet metal by actually melting the metal. High-pressure gasses, typically nitrogen or oxygen, are used along with the laser beam. The cutting head moves over the metal plate to create an exothermic reaction that delivers the precision cut details. Laser can also be used for non-through fabrication such as engraving, however, there are a few differences. The main difference between cutting, etching and engraving is the lens focal length. A laser-engraving machine has a shorter focal length, which results in high quality detail work. A laser cutter, on the other hand, uses a longer focal length, which allows for cutting thicker materials with a more precision cut and at faster speed.

Laser Cutting Process Pros/Cons

Pros

- Highly accurate and repeatable profiles
- No tooling required
- Low lead time
- Supports highly complex geometry
- Requires limited operator interaction during operation
- High sheet utilization (small kerf)
- High material versatility
- Can generate non-through sheet features (engraving)
- High speed in thinner material gages

Cons

- 2-D profiles only
- Accuracy and quality is operator dependent
- Sharpness of edge degrades as the material thickness increases
- Thicker materials may have a heat effected region where the material is cut

Manual Press Process

Manual Press Brake Process

A manual press brake is a machine used to make repeatable, precise, and reliable bends in metal sheets. They can be fitted with various types of punches and dies to control the bend of the sheet. Most come fitted with an automated back gauge that aids in positioning and holding the sheet, greatly increasing repeatability. Press brakes can produce either small parts or very long pieces with minimal tooling. They are useful in product fabrication from initial prototypes to low - medium production runs.

Manual Press Brake Process Pros/Cons

Pros

- Flexible process within fabrication window
- Low cost tooling
- Low lead time
- With custom tooling can support moderately complex geometry
- High material versatility

Cons

- Highly manual process
- Lower accuracy in comparison to a stage or progressive solution
- Accuracy and quality is operator dependent
- Complex part will require multiple bending operations (higher cost)

Single Stage / Transfer Die Process

Single Stage Fabrication Process

Single stage tooling (also referred to as manual dies) is a simpler process that relies on one stage, rather than multiple stage dies, to shape flat sheet metal to its predetermined specifications. During this tooling process, which can include punching, blanking, embossing, bending, forming, drawing, flanging, coining, and more, a part is either loaded by hand (manual dies) or loaded by machine tending robots (transfer dies). Despite being a single-stage operation, this process can effectively handle short- and long-volume runs. Moreover, precision metal stamping creates complex part geometries including three-dimensional parts with tight tolerances.

Single Stage Process Pros/Cons

Pros

- A cost-effective method for manufacturing
- Easy to setup
- Requires fewer adjustments
- A high-quality, single-stage press has an extensive lifespan
- Offers a less complex, straightforward process for the production of parts
- Allows for a quicker return on investment (ROI)
- Requires minimum capital investment

Cons

- It generally has a slower rate of production than progressive die stamping
- It can only handle one step at a time
- Dies need to be replaced when performing a new operation

Progressive Die Process

Progressive Die Process

Progressive die stamping is a cutting and cold-forming process that is frequently used to manufacture at high volume. The flat sheet metal roll is uncoiled and the strip stock is advanced to each stamping station, where it can be coined, punched, stamped, notched, bent, drawn, blanked, and more to shape the part to its predetermined specifications. After being processed through the stamping die, the finished part is removed from the strip stock. Essentially, progressive metal stamping is attractive because it’s automated and the machine does the work by following an assembly line of steps before achieving its final form.

Progressive Die Process Pros/Cons

Pros

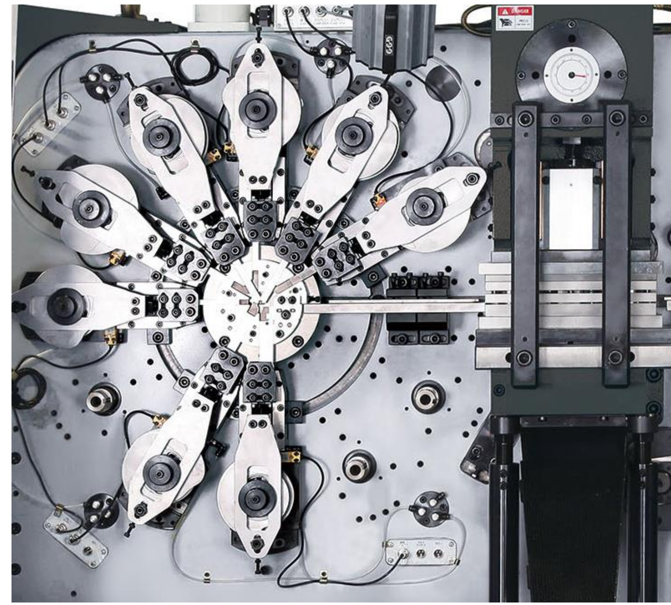
- Reduced labor costs
- Quick setup time
- Minimal scrap
- Requires only one press to make part
- Allows close to unattended machining operation on the plant floor
- High repeatability and reproducibility
- Rapidly, and consistently, produces multiple small parts with tight tolerances
- Scales to long production run lengths
- Decreased downtime
- Allows control of part location throughout the stamping process

Cons

- Higher cost tooling than stage or transfer dies
- Usually requires precision alignment and setup procedures
- A coil feeder system is needed
- An open-ended press is needed so that the metal can be fed into the die
- If one station is damaged, the entire die set needs to be removed

Multi Slide Process

Forming Stations   Punching Station



Multi Slide Process

The process of Multi-Slide Metal Stamping is a bit more complex than the other methods of metal stamping. In four-slide or multi- slide stamping the metal or wire is approached from multiple sides or angles by the dies, either successively or simultaneously. This produces a final product that is either coiled or bent, depending on the desired result and dies used. Four-slide technology is an excellent process for stamping and forming small, intricate parts with multiple bends from wire or thin strip metal. This process is used to produce springs, clips, brackets, etc.

Multi Slide Pros/Cons

Pros

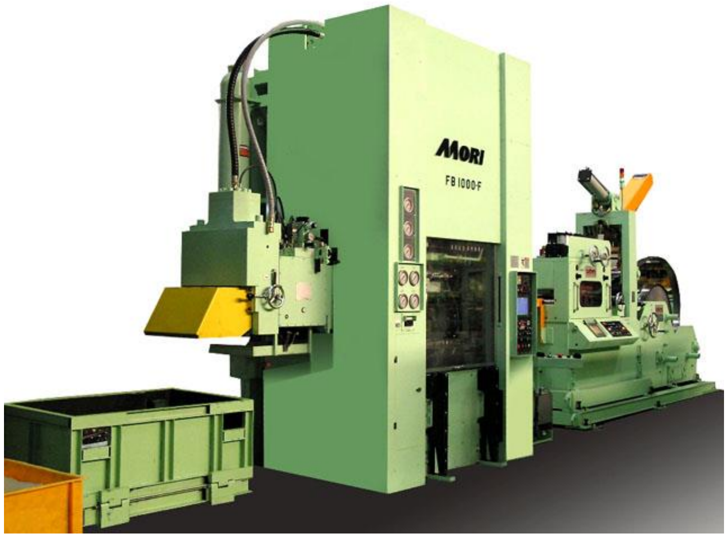
- Reduced labor costs
- Quick setup time
- Minimal scrap
- Requires only one press to make part
- Allows close to unattended machining operation on the plant floor
- High repeatability and reproducibility
- Rapidly, and consistently, produces multiple small parts with tight tolerances
- Scales to long production run lengths
- Decreased downtime
- Allows control of part location throughout the stamping process

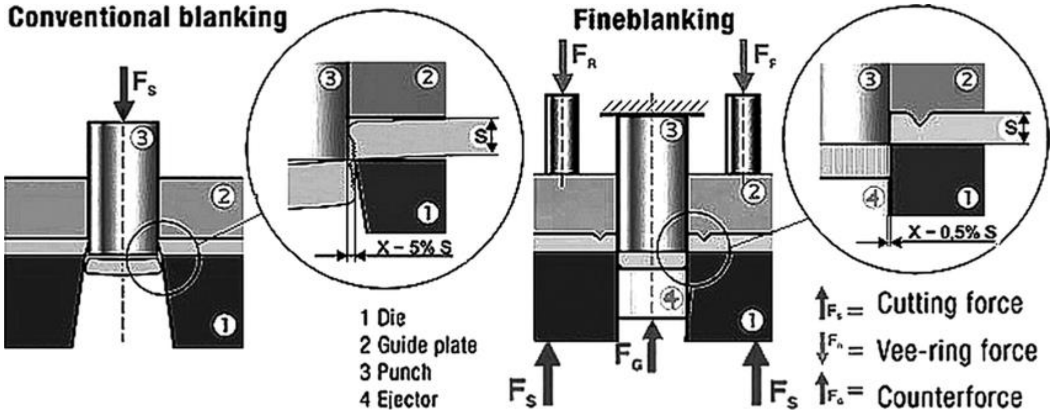
Cons

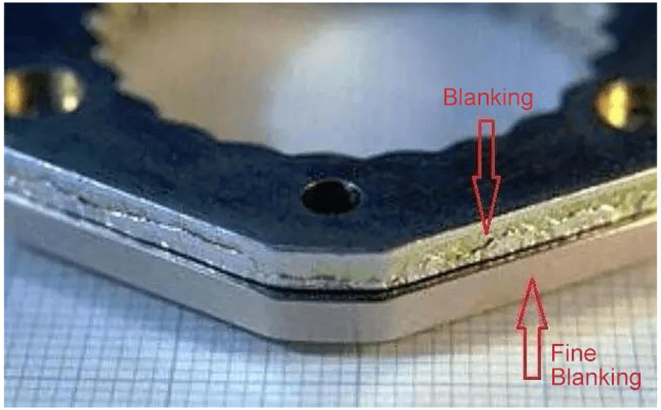
- Higher cost tooling than stage or transfer dies
- Usually requires precision alignment and setup procedures
- A coil feeder system is needed
- If one station is damaged, the entire die set needs to be removed

Fine Blanking Process

- Fine Blanking







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Fine Blanking Process

Fine blanking is a hydraulic metal stamping process that clamps both the top and bottom surfaces of the material during the forming process. This allows die clearance to be minimized providing parts with smooth edges and high accuracy. Because of the stability and precision of the process many secondary operations such as: grinding, milling, broaching, reaming, gear hobbing, forming, and shaving can potentially be eliminated. Fine blanking can produce, in a single step, a part that would require multiple operations, set-ups, and man-hours using other processes. As a result fine blanking is frequently the lowest total cost option for the appropriate component.

Fine Blanking Pros/Cons

Pros

- Extremely flat parts
- Improved edge quality, die brake is minimized
- Excellent dimensional control
- Excellent accuracy
- Excellent repeatability and reproducibility
- Minimizes the need to secondary finishing operations
- Flexibility to fabricate thick parts up to 17mm thick
- Supports variable part thickness

Cons

- Slower process speed compared to stamping
- Higher equipment investment
- Higher cost if secondary operations are required

Turret Press Process Capabilities

Description	Tolerance mm (inch)
FEATURE to FEATURE	+/-0.10 (0.004”)
PUNCHED feature size	+/-0.10 (0.004”)
FEATURE with multiple hits	+/-0.20 (0.008”)
FEATURE to EDGE	1.33T (material thickness)

Laser Cutting Press Process Capabilities

Description	Tolerance mm (inch)
FEATURE to FEATURE	+/-0.10 (0.004”)
FEATURE to EDGE	+/-0.10 (0.004”), 1.33T (material thickness)
Minimum KURF	+/-0.2 (0.008”)
Minimum FEATURE size	0.25T, 25% material thickness

Hand Brake Press Process Capabilities

Description	Tolerance mm (inch)
FOLD to FOLD	+/-0.5 (0.020”)
FOLD to FEATURE or EDGE	+/-0.4 (0.015”)
BEND angle	+/-1.5°
Minimum RADIUS	1.33T (material thickness)

Progressive Tool Process Capabilities

Description	Tolerance mm (inch)
Distance between FEATURES in the same station	+/-0.05 (0.002”)
Distance between FEATURES in different stations	+/-0.2 (0.008”)
Distance between FOLDS in the same station	+/-0.2 (0.008”)
Distance between FOLDS in different stations	+/-0.4 (0.016”), [+/-0.3 (0.012”)]
FOLD to FEATURE or EDGE	+/-0.25 (0.010”), [+/-0.1 (0.004”)]
Bend ANGLE	+/- 1°
Minimum HOLE diameter, SLOT width	1.33T (material thickness)
HOLE or FEATURE size	+/-0.08 (0.003”), [+/-0.03 (0.001”)]
Flatness	0.15/100mm, (0.006”/4”)
Straightness	0.12/100mm, (0.005”/4”)

[Fine tolerances achievable at additional cost]

Multi Slide Process Capabilities

Description	Tolerance mm (inch)
Distance between FEATURES	+/-0.05 (0.002”)
Distance between FOLDS	+/-0.15 (0.006”)
FOLD to FEATURE or EDGE	+/-0.15 (0.006”)
Bend ANGLE	+/- 1°
Minimum HOLE diameter, SLOT width	1.33T (material thickness)
HOLE or FEATURE size	+/-0.05 (0.002”)
Flatness	0.15/100mm, (0.006”/4”)
Straightness	0.12/100mm, (0.005”/4”)

Fine Blanking Process Capabilities

Description	Tolerance mm (inch)
Distance between FEATURES	+/-0.03 (0.001”)
HOLE or FEATURE size	+/-0.02 (0.001”)
Minimum HOLE Diameter or SLOT width (steel)	0.8T (material thickness)
Minimum HOLE Diameter or SLOT width (stainless steel)	1.0T (material thickness)
Flatness	1.0T (material thickness)

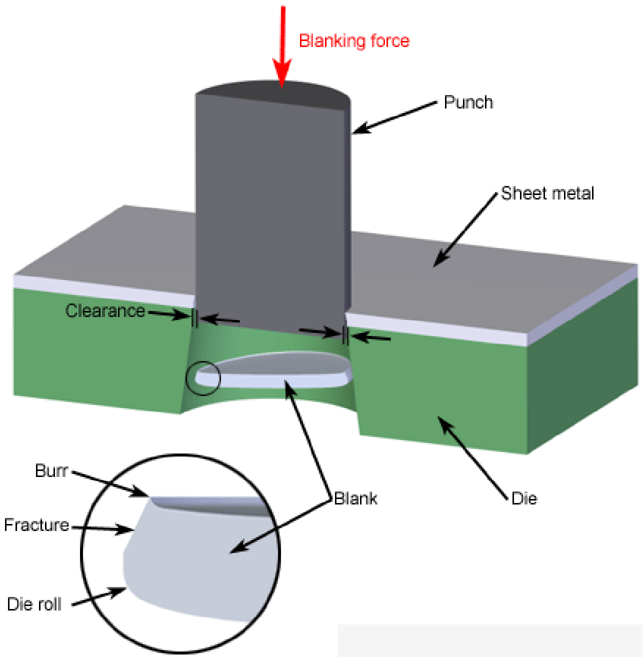
CHAPTER 3: SHEAR MECHANICS

Sheet Metal Mechanics

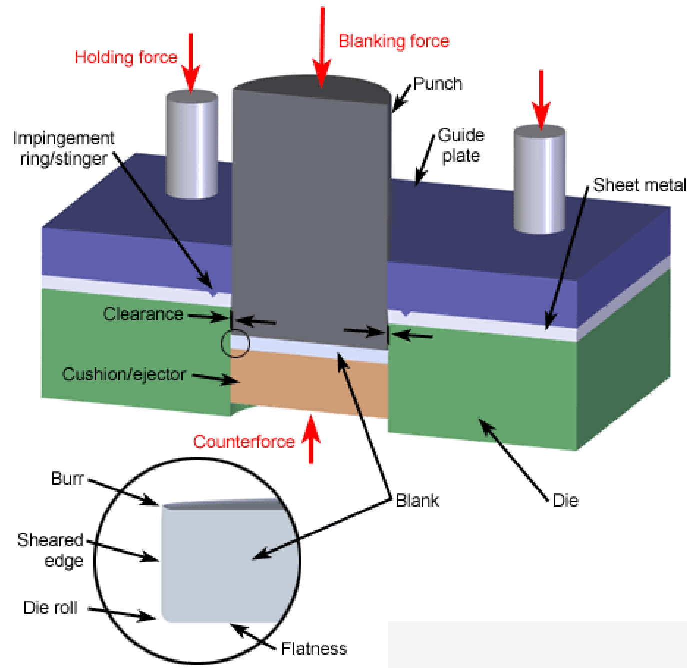
- Shear Mechanics
  - Conventional
  - Fine Blanking
  - Die Clearance
  - Tonnage Reduction Strategies
- Bending Mechanics and Methods
  - Mechanics
  - V Bending
    - Bottoming Bending
    - Coining
    - Air Bending
  - Roll Bending
  - Wipe Bending
  - Rotary Bending

Sheared Edge Characteristics

Punching

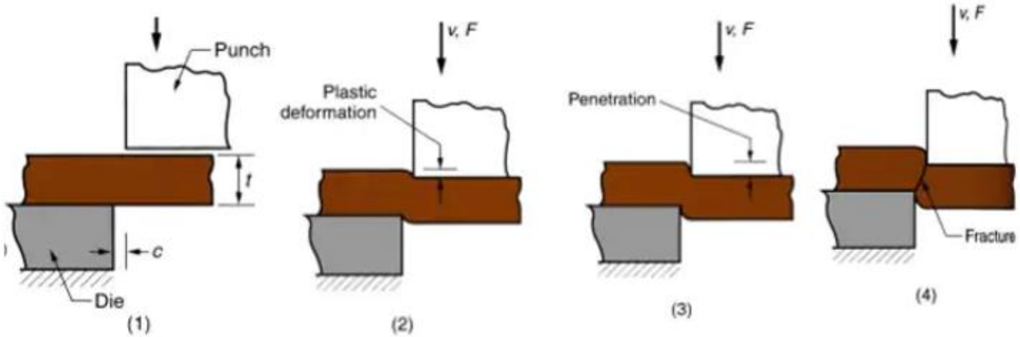


Fine Blanking

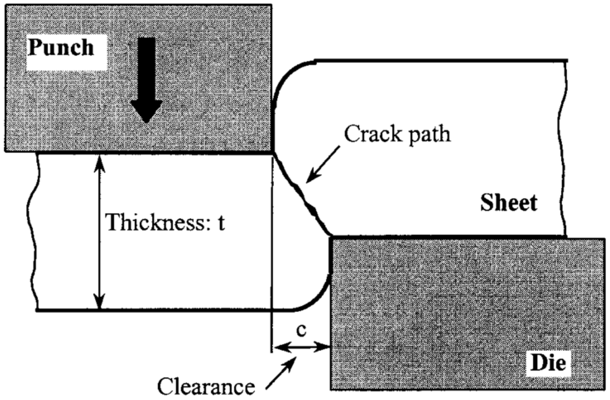


Sheared Edge Characteristics Conventional Punched Feature (cont.)

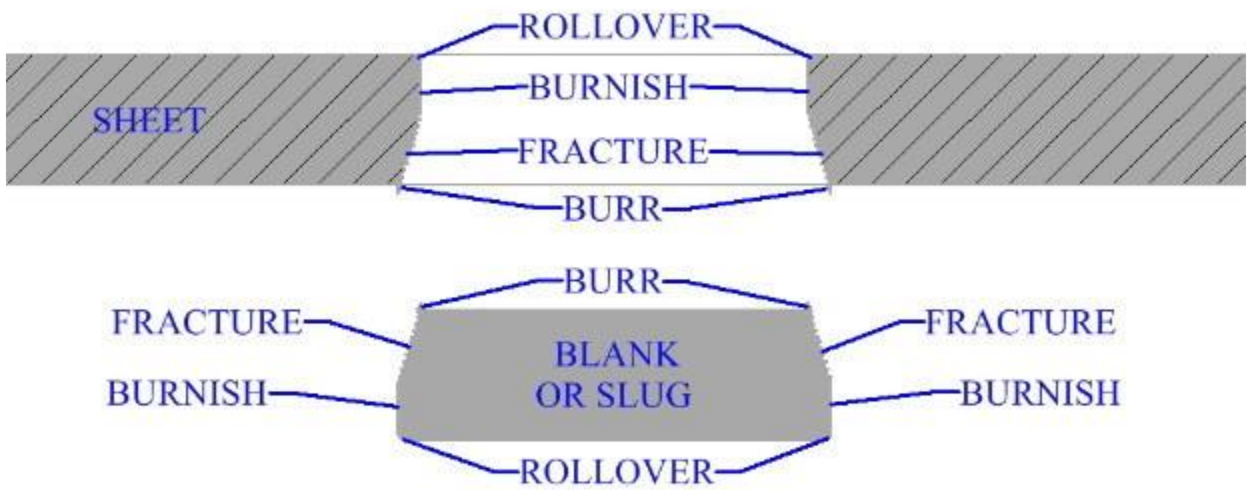
Sheet Metal Cutting - Shearing



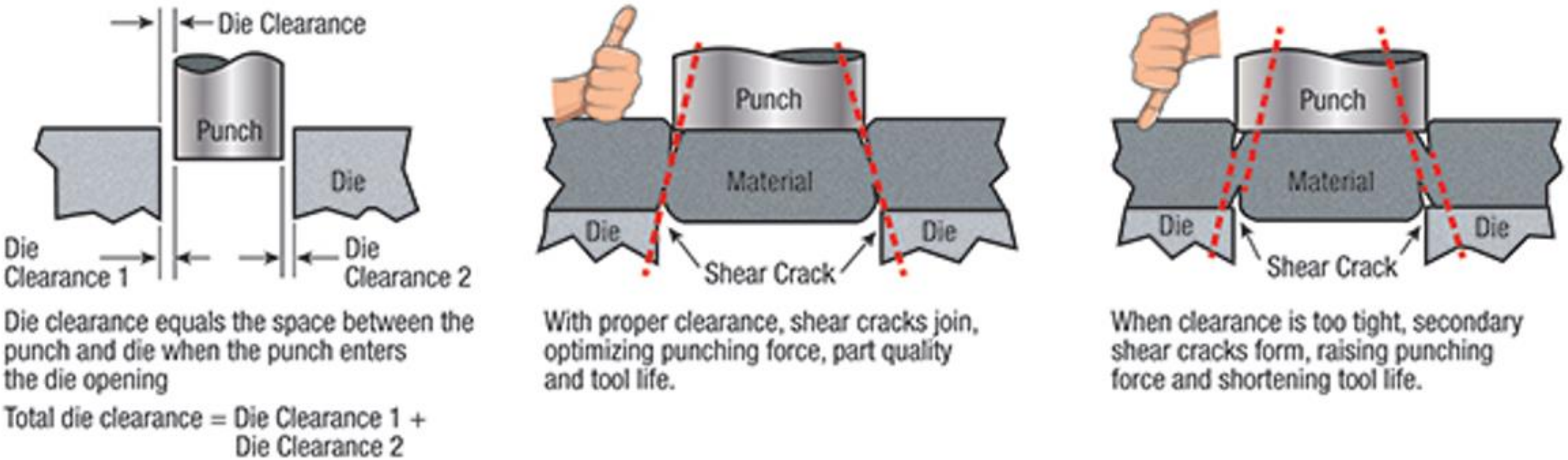
Shearing of sheet metal between two cutting edges:
(1) just before the punch contacts work;
(2) punch begins to push into work, causing plastic deformation; Shearing of sheet metal between two cutting edges:
(3) punch compresses and penetrates into work causing a smooth cut surface;
(4) fracture is initiated at the opposing cutting edges which separates the sheet.



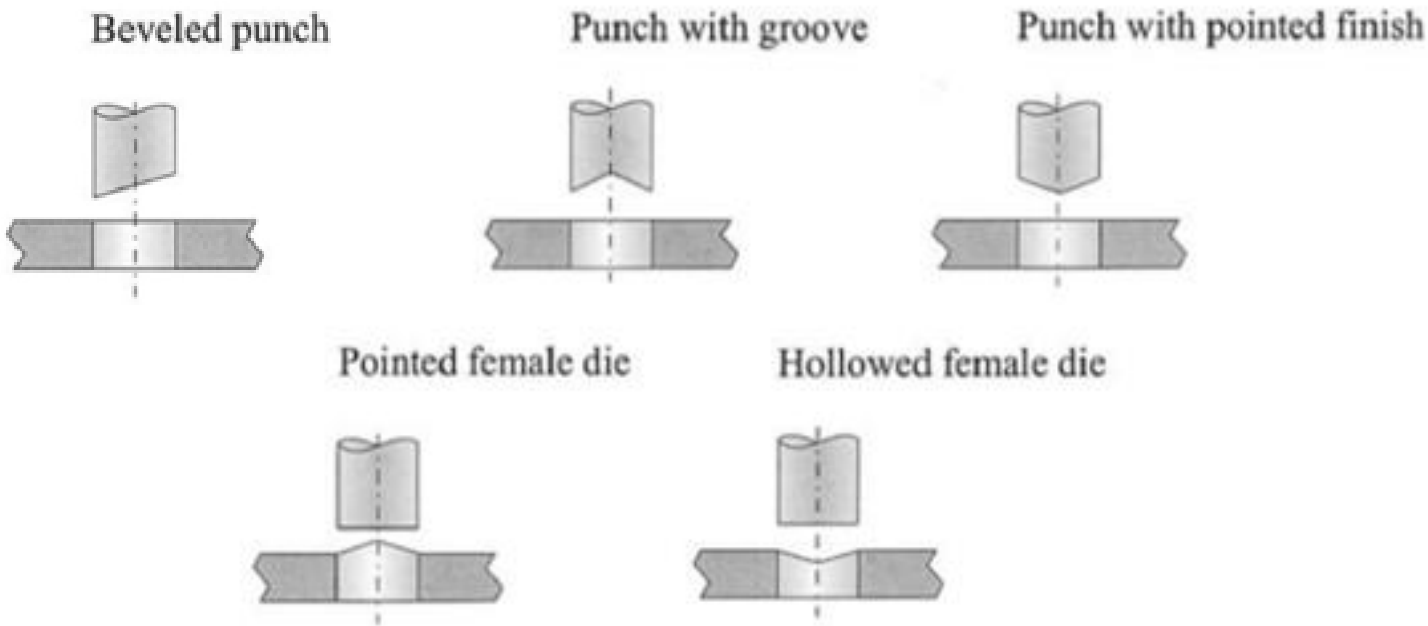
Sheared Edge Characteristics Conventional Punched Feature (cont.)



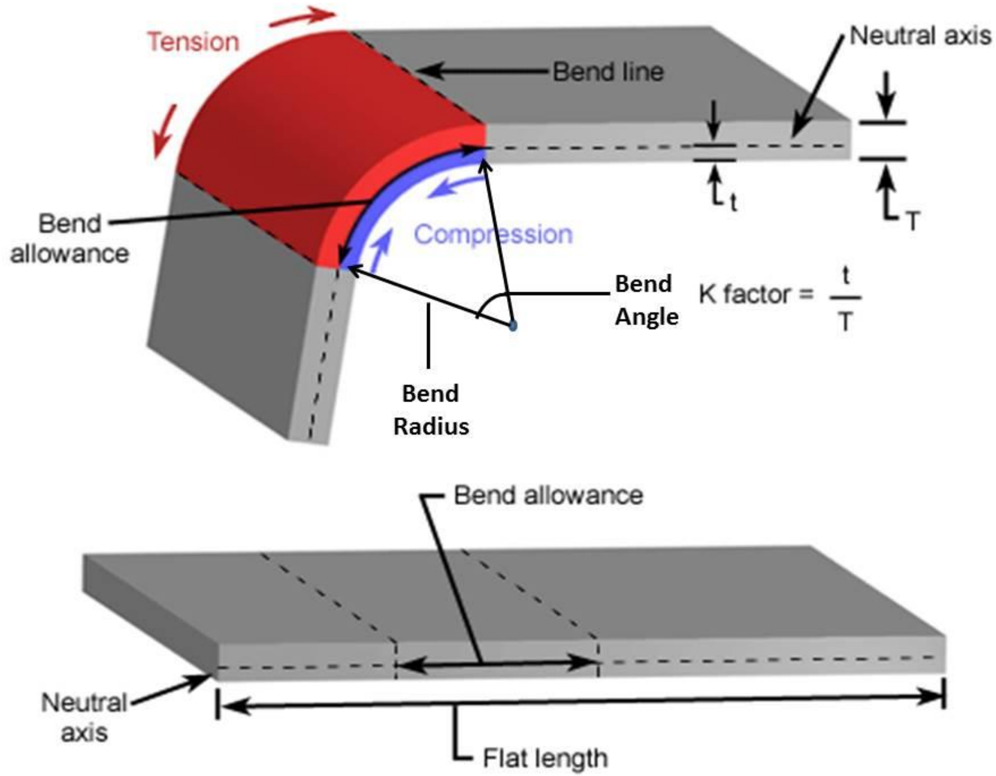
Shear Dynamics Based on Die Clearance



Methods to Reduce Press Tonnage with Punch Geometry

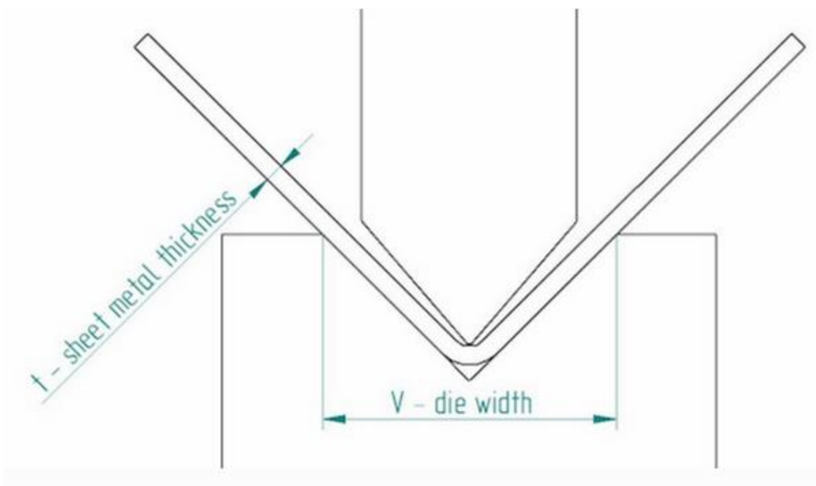


Bending Mechanics



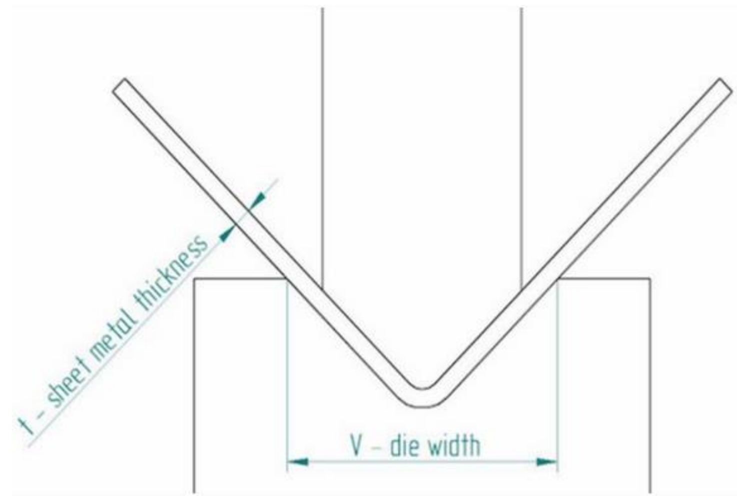
Bottoming Bending

Bottoming or bottom bending involves compressing the sheet metal to the preset die's bottom to form a defined angle and shape. In bottoming, the position and shape of the die angle determine the outcome of the bend. Also, the spring-back of the compressed sheet metal is impossible. This is because the force of the punch and the die's angle conforms the sheet metal to a permanent structure (material thinned at base of bend to coin the bend angle)



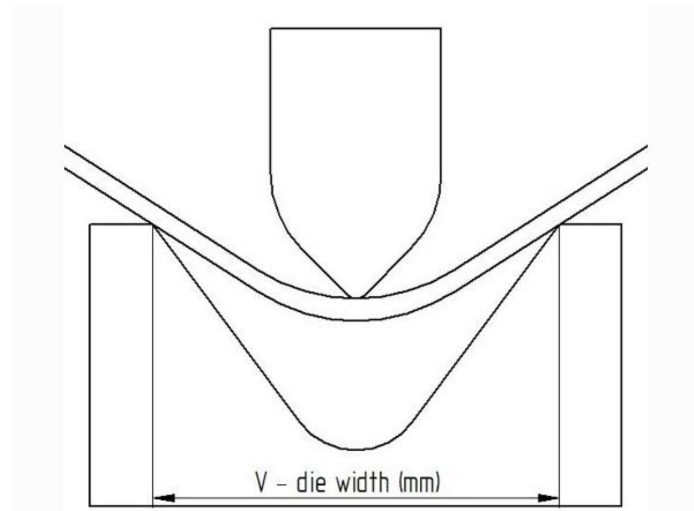
Coin Bending

Coining is a bending method widely employed for its precision. In the process, there is no spring-back of the sheets. This is because the coin penetrates the sheet metal at a small radius, creating a dent present on a coin.



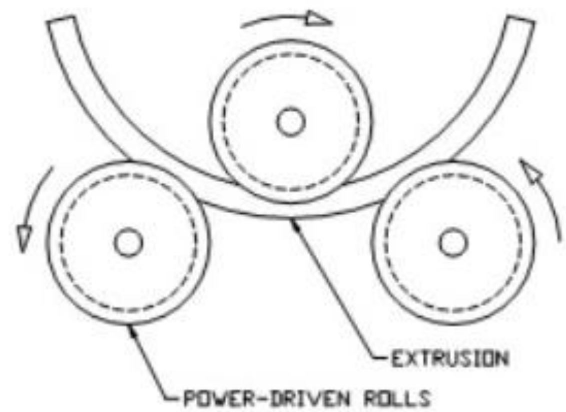
Air Bending

In air bending, the punch exerts a force on the sheet metal resting on both points of the die's opening. A press brake is usually employed during the V bending as the sheet metal does not come in contact with the bottom of the die. Air bending or partial bending is a less accurate method compared to bottoming and coining. Lastly air bending is the only method prone to the spring-back of the sheet metal.



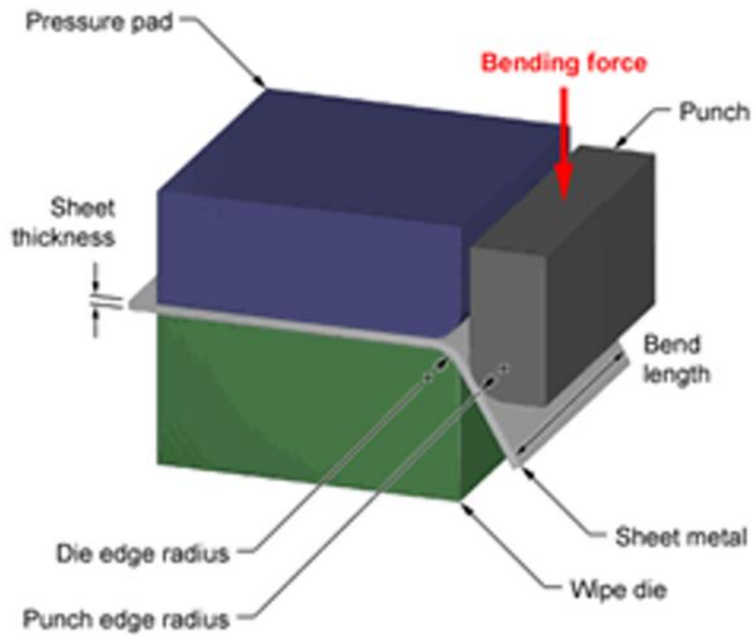
Roll Bending

Roll bending is a method used to bend sheet metals into rolls or curved shapes. The process employs a hydraulic press with three sets of rollers to make different bends or a big round bend. It is useful in forming cones, tubes, and hollow shapes as it takes advantage of the distance between the rollers to make bends and curves.



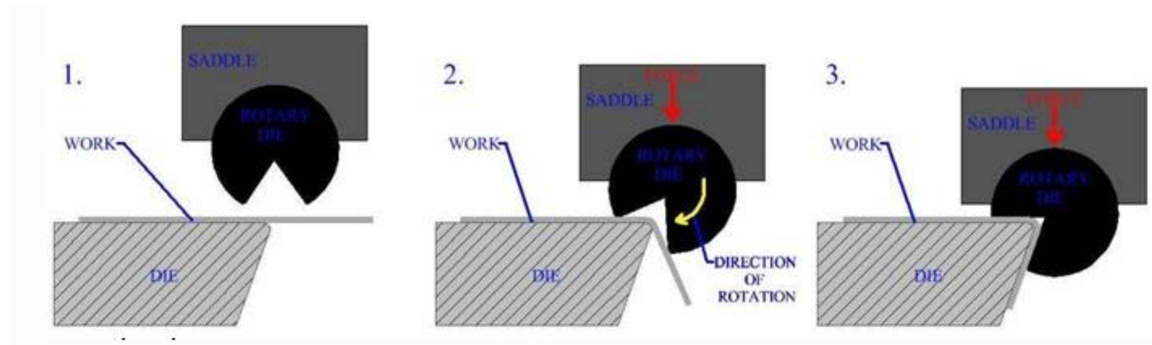
Wipe Bending

Wipe bending requires a **pressure pad to hold the sheet metal parts against the die**. The punch then moves up and down to bend the edge of the sheet metal. Wipe bending is much faster than any other bending process, but there is always a risk of scratches because the tool is moving over the surface directly.



Rotary Bending

Rotary bending has an advantage over wipe bending or V-bending because it does not scratch the material's surface. It is also ideal because it can bend materials into sharp corners. For example, it is used in bending corners greater than 90^0.



CHAPTER 4: MATERIALS

Materials

- Specifications
- Thickness
- Coatings
- Laminated materials

Material Specifications Steel

- SPCC – commercial grade Basic CRS, general use, dead mild steel
- SPCD - drawing quality CRS, general use
- SPCE - deep drawing quality CRS
- SECC – commercial stamping grade CRS, electro-galvanized zinc layer on the surface
- SECD - drawing grade CRS, electro-galvanized zinc layer on the surface
  SECE - drawing grade CRS, electro-galvanized zinc layer on the surface, improved thermal conduction
- SGCC - commercial grade CRS, hot-dipped galvanized
- SGCD - drawing grade CRS, hot-dipped galvanized
- SPTE - thin low-carbon, tin plated, used in the canning industry

Thickness Specifications Stainless Steel

Thickness (mm)	Tolerance mm , [“]
0.6 – 0.79	+/-0.05 [0.002”]
0.80 – 0.99	+/-0.055 [0.0022”]
1.0 – 1.19	+/-0.06 [0.0024”]
1.20 – 1.49	+/-0.07 [0.0028”]
1.50 – 1.99	+/-0.08 [0.0032”]
2.0 – 2.49	+/-0.09 [0.0035”]
2.5 – 2.99	+/-0.11 [0.0043”]
3.0 – 3.00	+/-0.13 [0.0051”]

- Specification are for commercial quality (CQ)
- Material usually delivered on the minus side of the tolerance

Thickness Specifications SGCC Steel

Thickness (mm)	Tolerance mm , [“]
0.40 – 0.60	+/-0.05 [0.0020”]
0.61 – 0.80	+/-0.06 [0.0024”]
0.81 – 1.00	+/-0.07 [0.0028”]
1.01 – 1.20	+/-0.08 [0.0032”]
1.21 – 1.60	+/-0.11 [0.0043”]
1.61 – 2.00	+/-0.14 [0.0055”]
2.01 – 2.50	+/-0.16 [0.0063”]
2.51 – 3.00	+/-0.19 [0.0075”]

- Specification are for commercial quality (CQ)
- Material usually delivered on the minus side of the tolerance

Thickness Specifications SECC Steel

Thickness (mm)	Tolerance mm , [“]
0.40 – 0.60	+/-0.03 [0.0012”]
0.61 – 0.80	+/-0.04 [0.0016”]
0.81 – 1.00	+/-0.05 [0.0020”]
1.01 – 1.20	+/-0.06 [0.0024”]
1.21 – 1.60	+/-0.08 [0.0032”]
1.61 – 2.00	+/-0.10 [0.0040”]

- Specification are for commercial quality (CQ)
- Material usually delivered on the minus side of the tolerance

Thickness Specifications Aluminum

Thickness (mm)	Tolerance mm , [“] Group I	Tolerance mm , [“] Group II
0.6 – 0.80	+/-0.03 [0.0012”]	+/-0.04 [0.0016”]
0.81 – 1.00	+/-0.04 [0.0016”]	+/-0.05 [0.0020”]
1.01 – 1.20	+/-0.04 [0.0016”]	+/-0.05 [0.0020”]
1.21 – 1.50	+/-0.05 [0.0020”]	+/-0.07 [0.0028”]
1.51 – 1.80	+/-0.06 [0.0024”]	+/-0.08 [0.0032”]
2.01 – 2.50	+/-0.07 [0.0028”]	+/-0.10 [0.0040”]
2.51 – 3.00	+/-0.08 [0.0032”]	+/-0.11 [0.0043”]
3.01 – 3.50	+/-0.10 [0.0040”]	+/-0.12 [0.0047”]

Group I = 1000 series, 3000 series, 4006, 4007, 5005, 5050 and 8011A

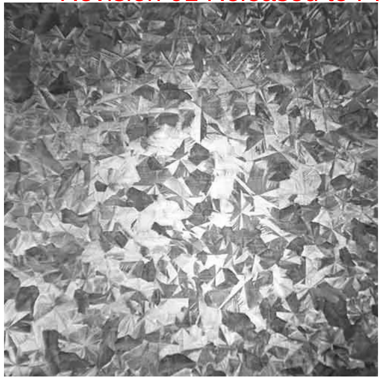
Group II = 2000 series, 6000 series, 7000 series, 3004, 5040, 5049, 5251, 5052, 5154A, 5454, 5754, 5182, 5083 and 5086

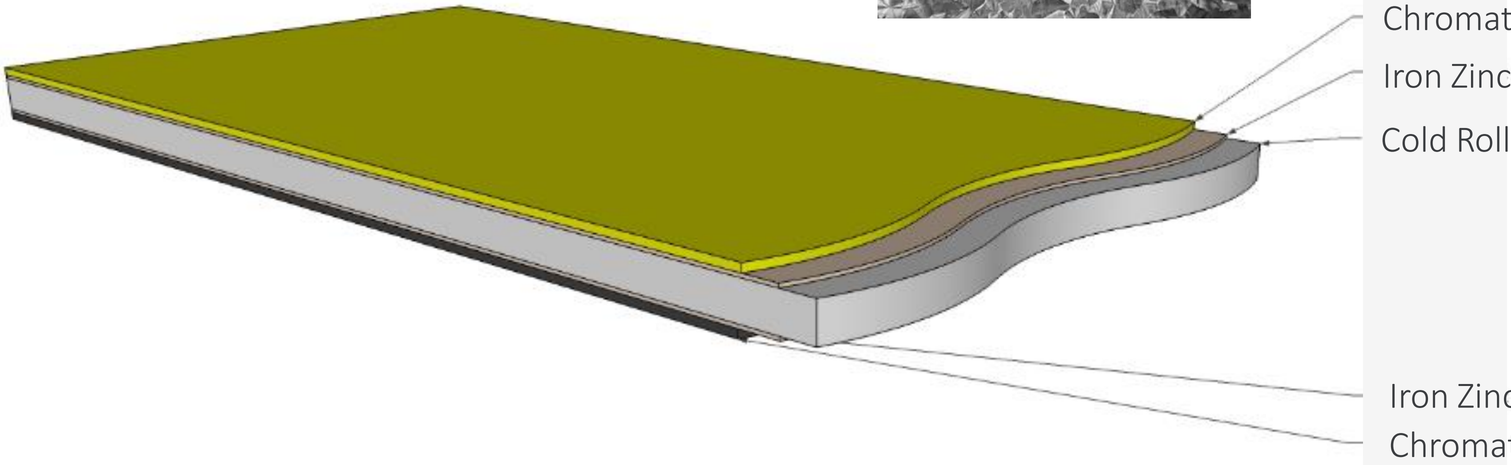
- Specification are for commercial quality (CQ)
- Material usually delivered on the minus side of the tolerance

Thickness Specifications US vs Metric

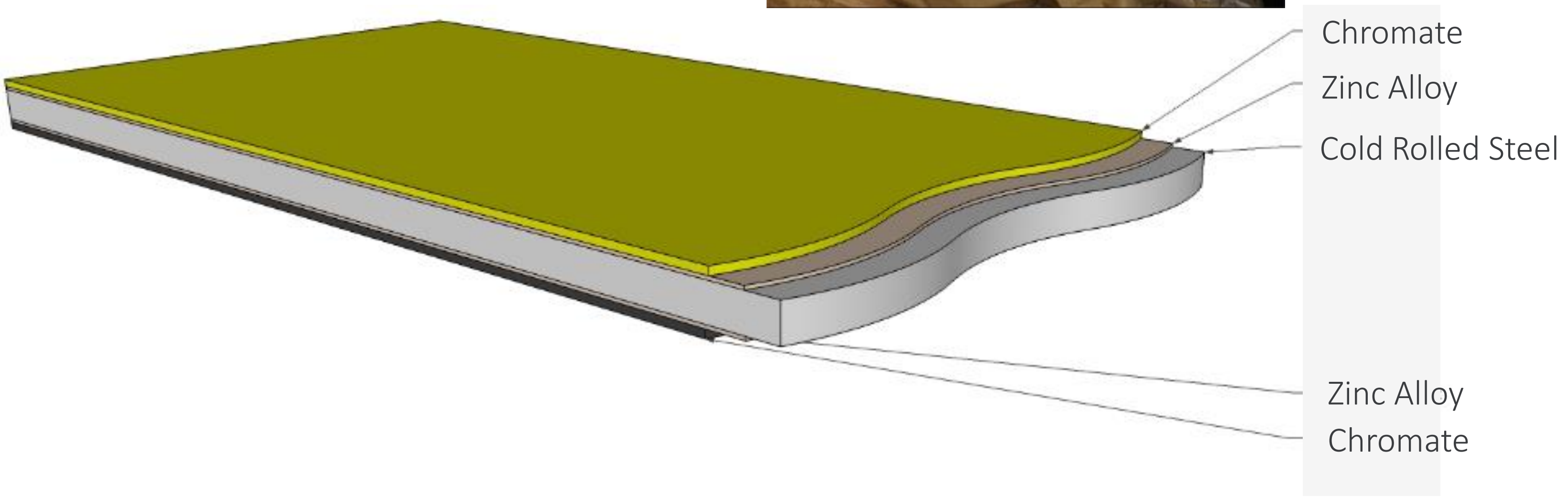
Gage	Millimeters	Inches
26	0.455	0.0179
24	0.607	0.0239
22	0.759	0.0299
20	0.912	0.0359
18	1.214	0.0478
16	1.519	0.0598
14	1.897	0.0747

Coating SGCC





Coating SECC



CHAPTER 5: DFX GUIDELINES

General Design Guidelines Checklist

- Specify punch direction
- Specify grain direction
- Specify areas that need deburring or coining
- Identify features that need to be grouped in a single station for best tolerancing
- Avoid secondary operations if possible
- Keep embosses and draws shallow
- Use standard readily available materials
- Use precoated materials, avoid unplated materials or secondary protective films
- Review design for handling and shipping damage
- Review strip layout before releasing hard tooling to the supplier
- Communicate features that are not stable during tooling so mitigation can be put in place

General Design Guidelines Checklist (cont.)

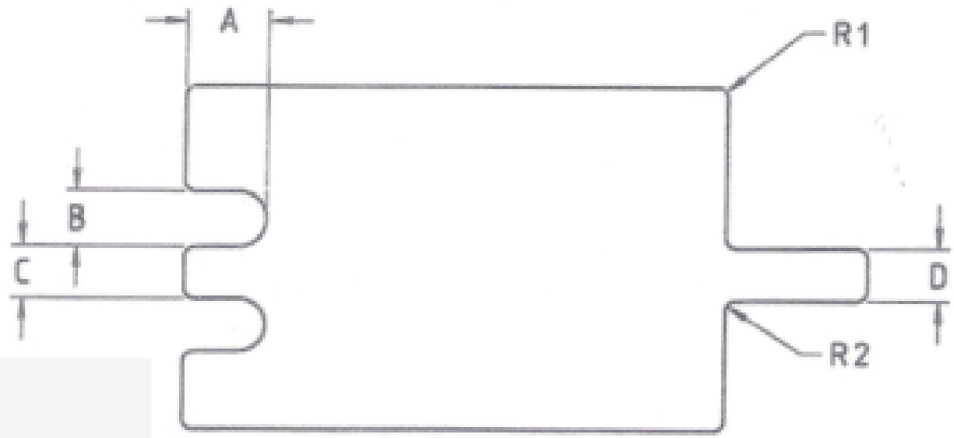
- Add tooling hole(s) to supply consistent reference features if possible
- Specify acceptable locations for carrier tabs
- Datum structure(s) aligns with the use and function of the part
- GD&T clearly defines features
- Ensure that process control dimensions can be measured at the press with simple tools
- Use sheared edges for all critical reference features
- Define maximum and minimum bend radius
- Design inspection fixtures and process in parallel with the part design
- Clearly specify critical and major dimensions

Guidelines to Minimize Secondary Operations

Secondary Operation	Option(s)
Tapped Hole	Extruded hole and Taptite
Threaded Hardware	Extruded hole and Taptite
Spacers	Bridge feature or emboss
Standoffs	Bridge feature or emboss
Spot Welds	Tox/Tog-L-Loc, Swaging
Rivets	Tox/Tog-L-Loc, Swaging
Screws	Tox/Tog-L-Loc, Swaging

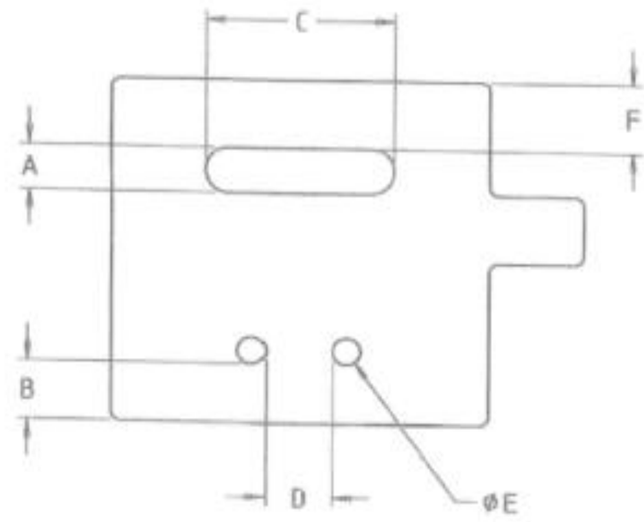
Design Guidelines

- A max = 2B or 6T
- B min = 1.33T
- C min = 2T
- D min = C
- R1 min = 0.5T or 0.4
- R2 min = 0.5T or 0.4



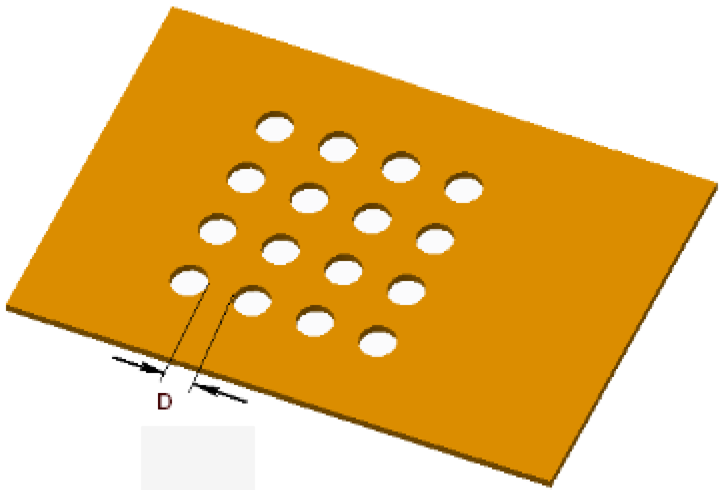
Design Guidelines

- A min = 2.0T
- B min = 2.0T
- C max = 50mm when A = minimum
- C min = 2R + T
- D min = 2T
- E min = 1.2T
- F min = 3T if C <= 10T
  4T if C > 10T



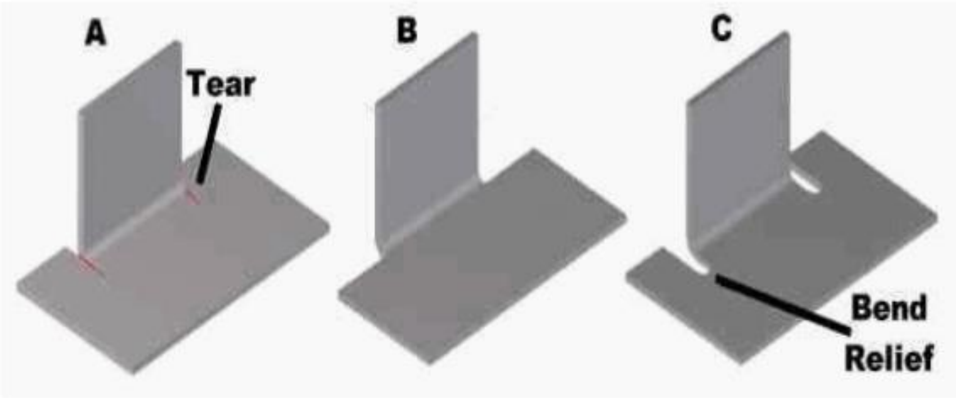
Design Guidelines

- Minimum distance between holes D = 1.2T

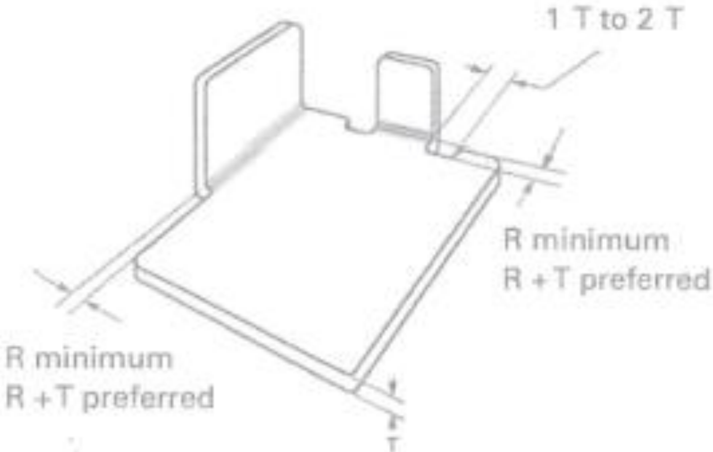


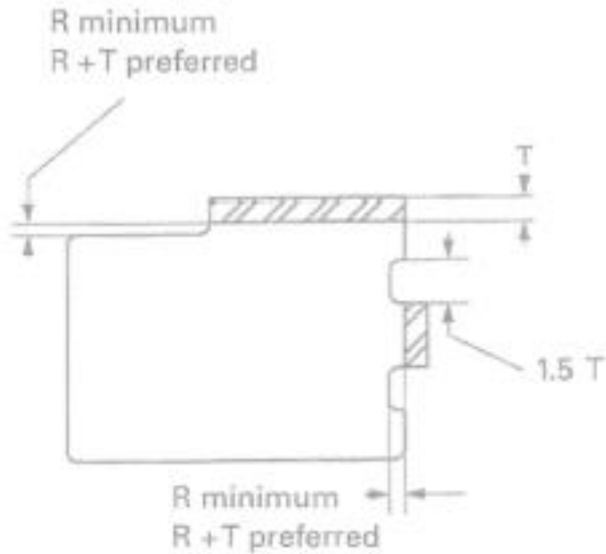
Bend Relief

Bend Relief - When a bend is made close to an edge the material may tear unless bend relief is given. Figure "A" shows a torn part. Figure "B" shows a part with the edge a sufficient distance from the form. This distance should never be less than the radius of the bend. Figure "C" shows a bend relief cut into the part, again the depth of the relief should be greater than the radius of the bend. The width of the relief should be a material thickness or greater.



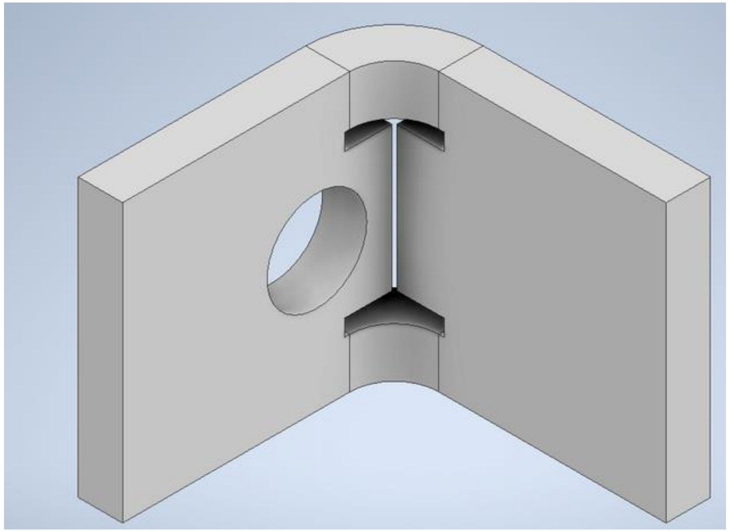
Bend Relief


1 T to 2 T
R minimum
R +T preferred
R minimum
R +T preferred
T

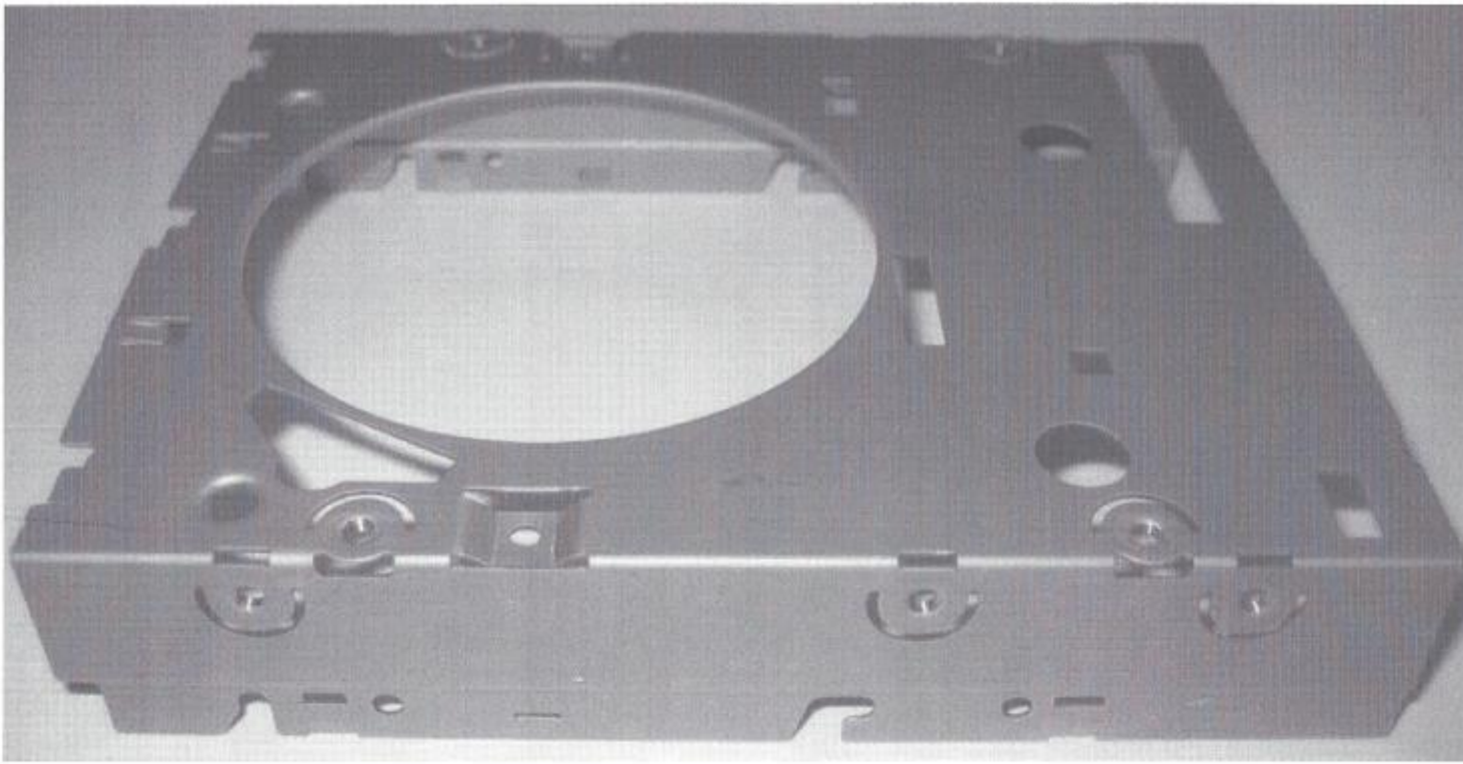

R minimum
R +T preferred
T
1.5 T
R minimum
R +T preferred

Bend Relief

Punched slot in bend allows hole to be located closer to the fold without deformation from the bending operation

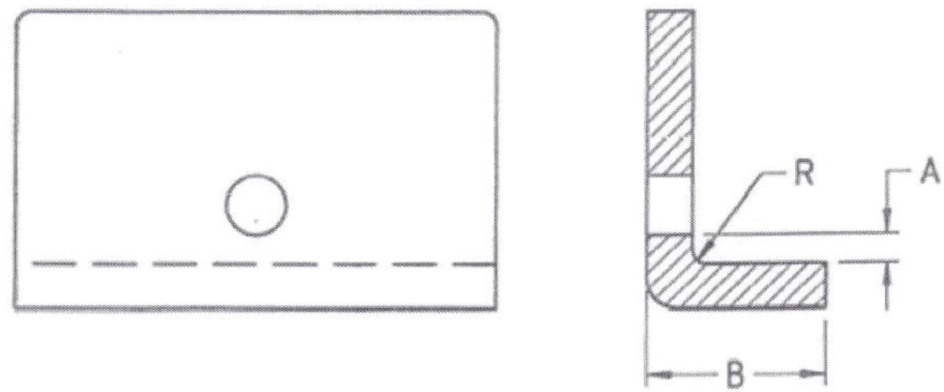


Bend Relief



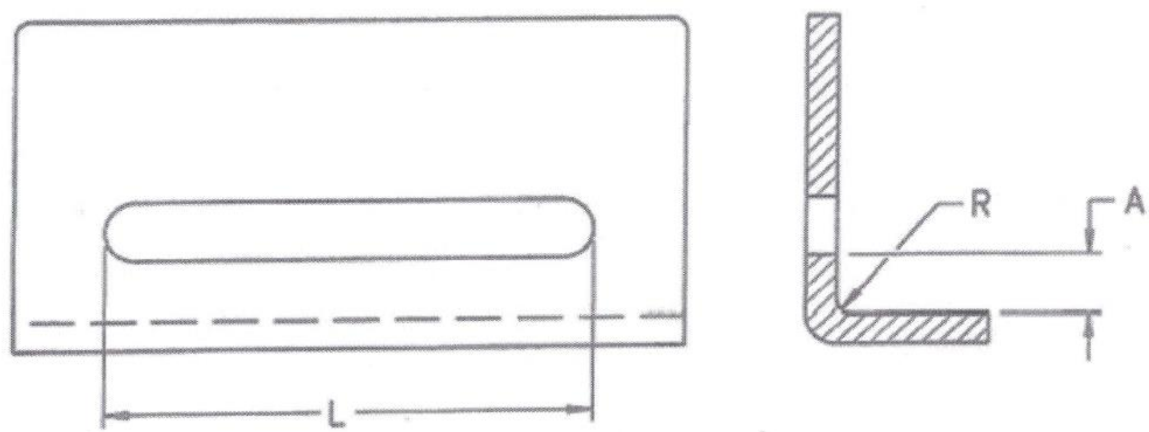
Hole Distance from Bend

- A min = 2.0T + R
- B min = 3T + R
- R min = 1.0T or 0.6 mm minimum



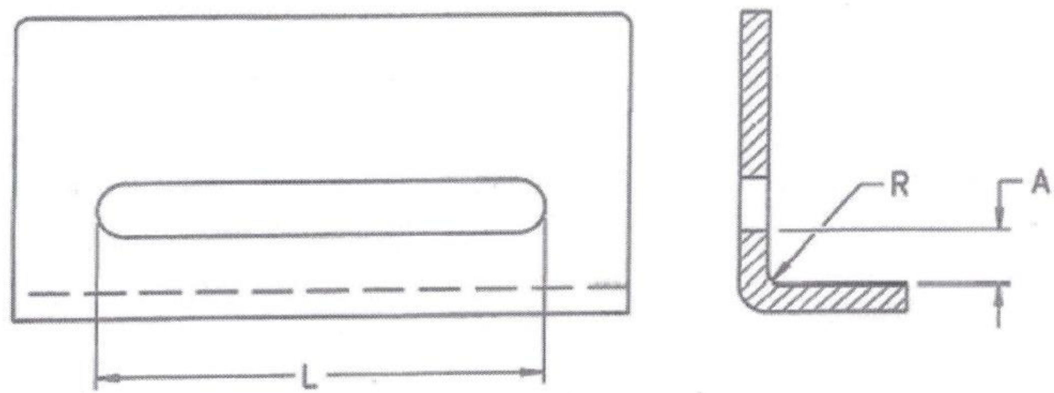
Slot Distance from Bend

- L < 50mm, A = 3T + R
- L > 50mm, A = 4T + R
- R min = 1.0T or 0.6 mm minimum

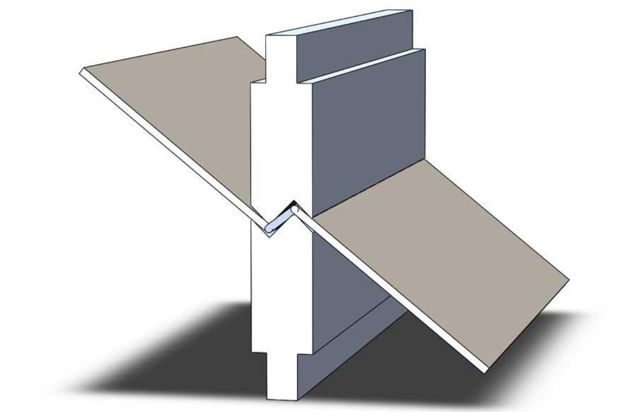


Bend Radius

- R min = 1.0T or 0.6 mm minimum
- When required (only when using low carbon steel R min = 0.5T

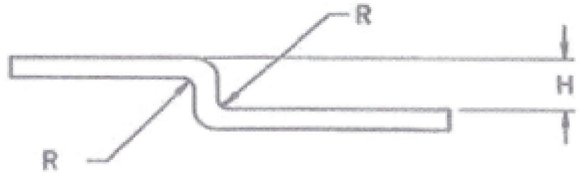


Z Bends



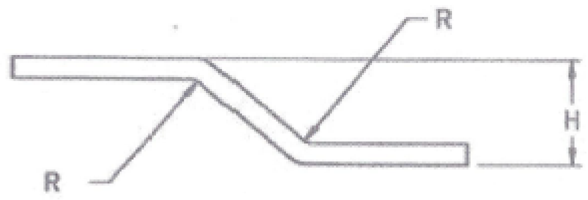
H <= 5T
R = 1.5T

90° Offset

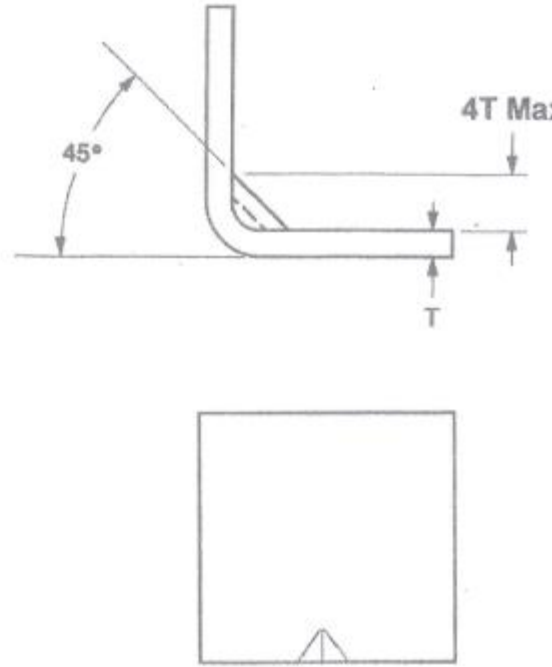


R = 0.2mm

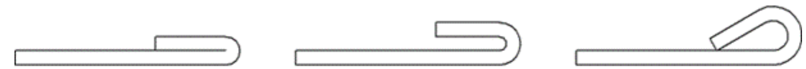
45° Offset



Stamped Gussets

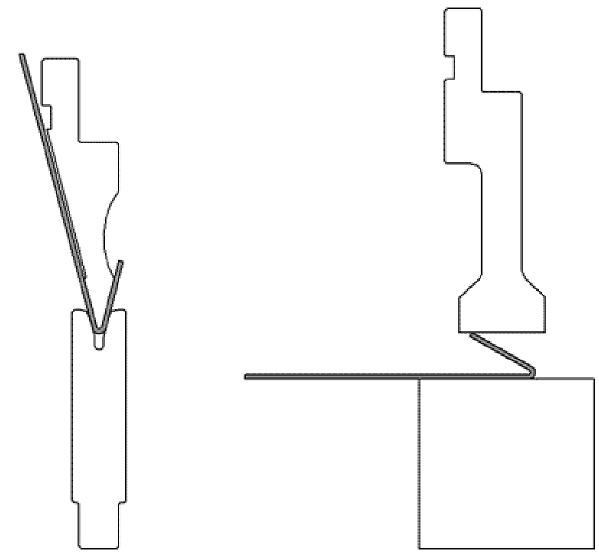


Hems



FLAT HEM | OPEN HEM | TEAR DROP HEM

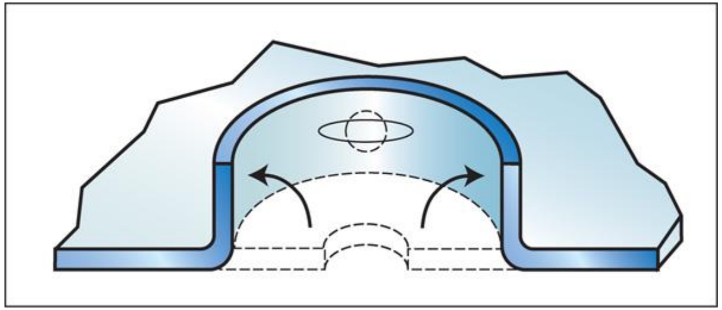
- Used to hide burrs, not improve flatness
- Operation requires multiple steps
- Tear drop is most common
- Minimum return flange height = 4T
- Minimum distance from hole = 2T
- Minimum distance from internal bend = 5T
- Minimum distance from external bend = 8T
- Flat hems tend to fracture and cause entrapment of fluids



Step 1, Acute angle bend

Step 2, flattening bar

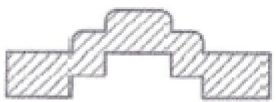
Extrusions

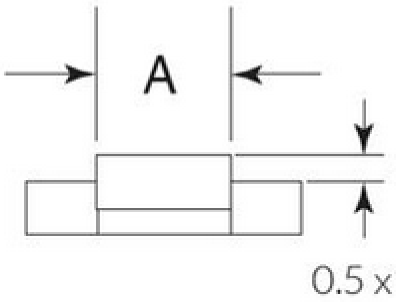


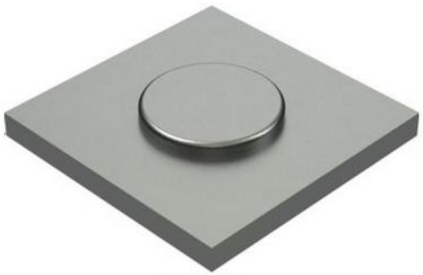
- Minimum distance between extrusions = 6T
- Minimum distance extruded hole to edge = 3T + 2R
- Minimum distance extruded hole to a bend = 3T + R
- Minimum extrusion height H should allow 4 threads of engagement (~80% strength achieved with 4 threads)
- H = 2.0T – 2.5T for a M3 screw
- H = 2.5T – 3.0T for a M4 screw

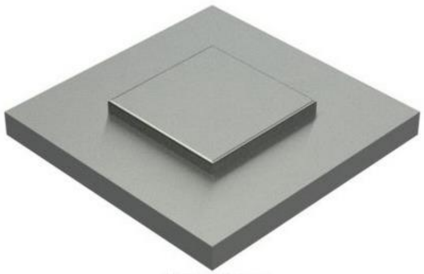
Half Shear

- Maximum height of single shear 0.5T
- Minimum width/diameter of feature ("A") 2T
- Multiple shears can be used to exceed total height of 0.5T





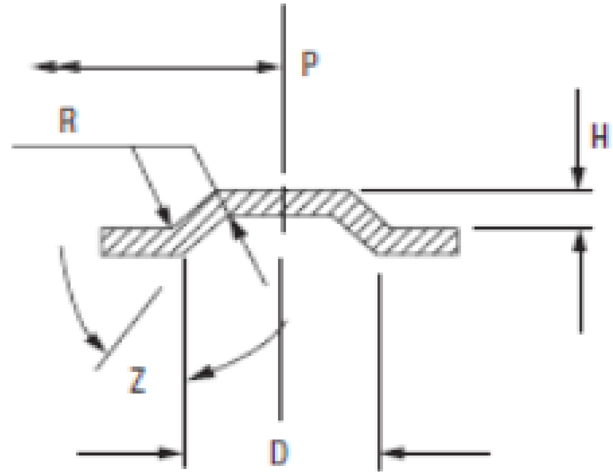


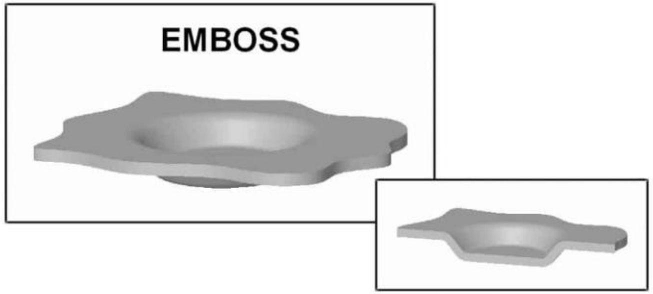


Dimples

- All distances are measured from the edge of the dimple radius
- Maximum diameter D = 6T
- Maximum height H = 4T
- Minimum height H = D * 0.3
- Minimum distance to Hole = 3T
- Minimum distance between Dimples = 4T + D
- Minimum distance Dimple to Bend = 2T
- Minimum distance Dimple to an part Edge P = 3T + D/2
- Angle Z >= 30°

ROUND EMBOSS TOOL

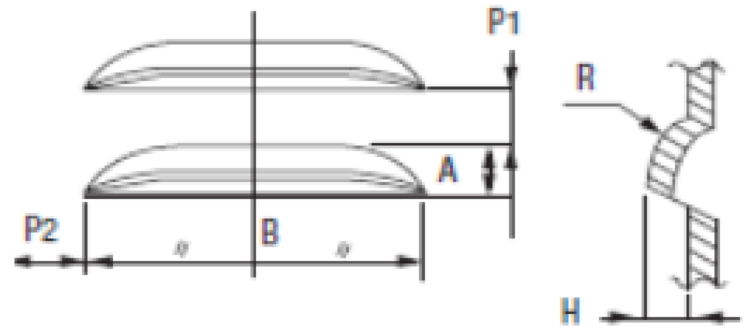




Section View

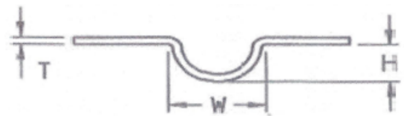
Louvers

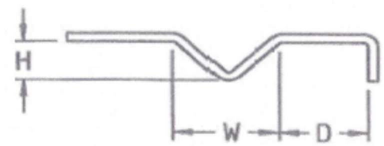
- Material thickness (T) range 0.8mm – 6 mm
- Minimum spacing (short edge) P1 = 5 mm
- Minimum spacing (long edge) P2 = 8 mm
- Maximum Height H = A * 0.6
- Minimum height H = 3T
- Minimum radius R > H



Embossments / Stiffing Rib

- Round
  - Maximum height H = 3T
  - Minimum width W = 3T
- V embossment
  - Maximum height H = 3T
  - Minimum width W = 3H
- Flat
  - The maximum depth = internal radius plus the external radius
- Minimum distance between parallel embossments = 10T
- Minimum distance to edge = 4T
- Minimum distance to Hole = 3T
- Minimum distance to Fold D = 4T

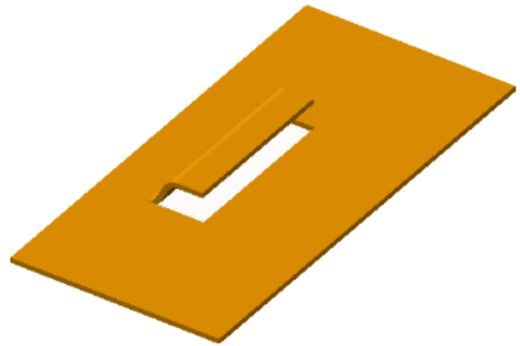


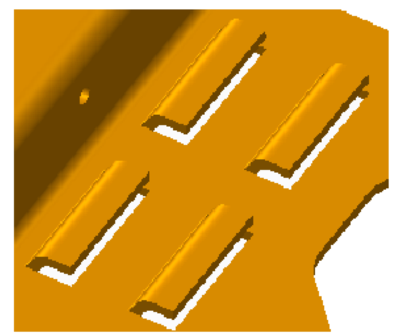


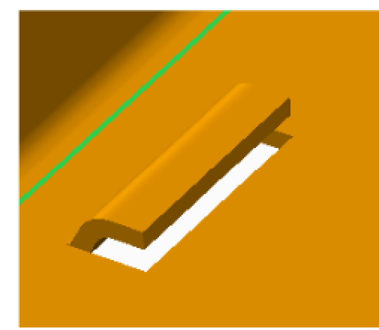
Flat

Lance and Form

- Minimum height H = 2T
- Minimum distance between lance and fold = 3T + R (R = inside bend radius)
- Minimum distance between lance and Hole = 3T
- Minimum web distance between lances = 6T
- No bend relief required

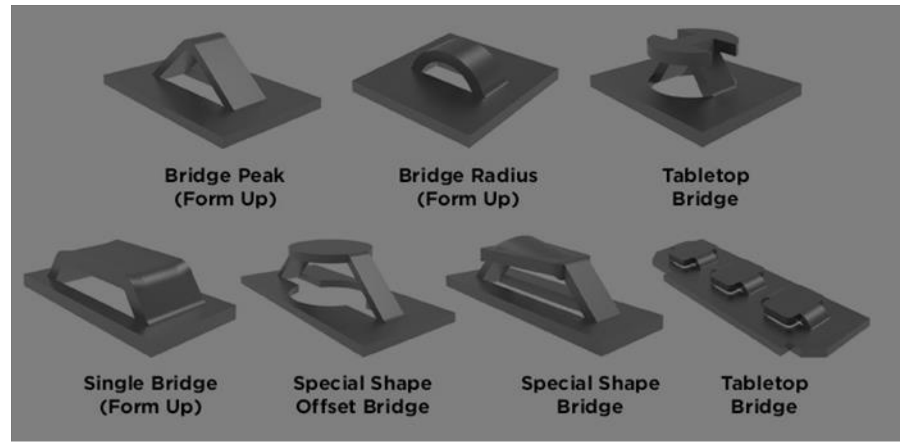






Lance Bridges

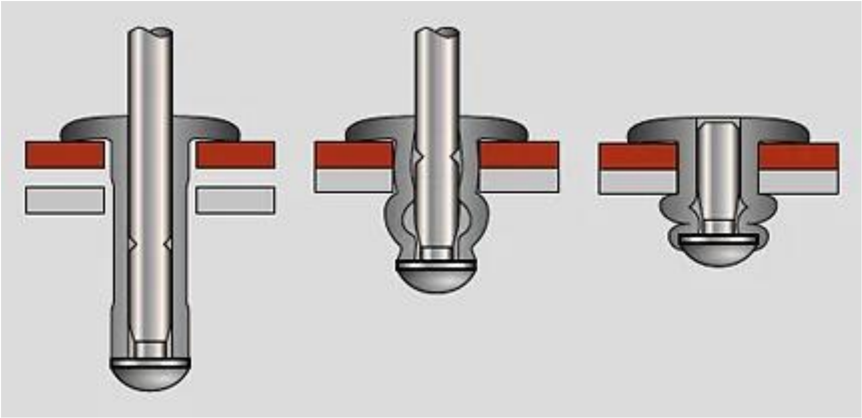
- Minimum height = 2T
- Maximum height = 5T
- Minimum distance between lance and fold = 3T + R (R = inside bend radius) highly dependent on tooling design, recommended min = 8T
- Minimum distance between lance and Hole = 3T
- No bend relief required

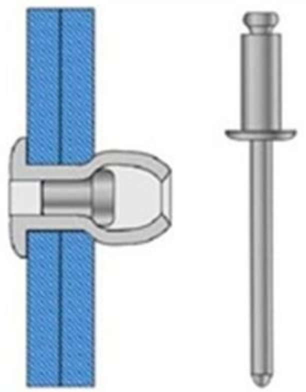


CHAPTER 6: JOINING TECHNIQUES

Blind Rivets   Pros / Cons

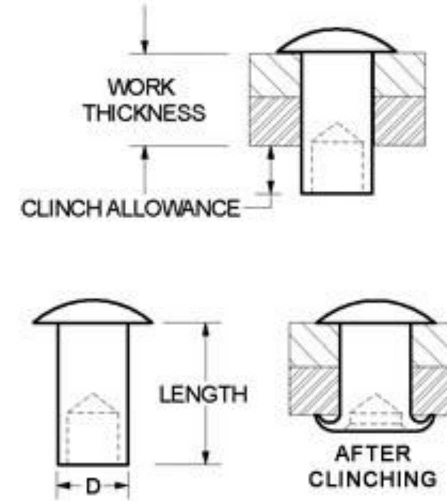
- Pros
  - Installed from one side
  - Installation tool is portable
  - Can be installed when only one side of the interface is accessible
  - Fast installation time
  - Low operator training required
  - Joining dissimilar materials
  - Low equipment cost
  - No damage to surface finish
- Cons
  - Back side cinch is not consistent
  - Low resistance to vibration and impact

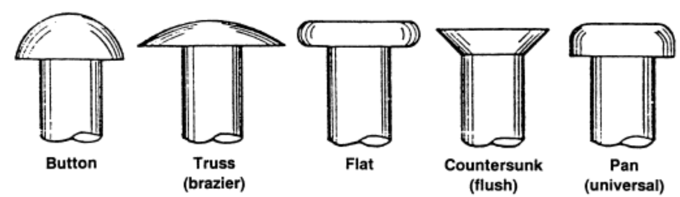




Hollow (Tubular) Rivets Pros / Cons

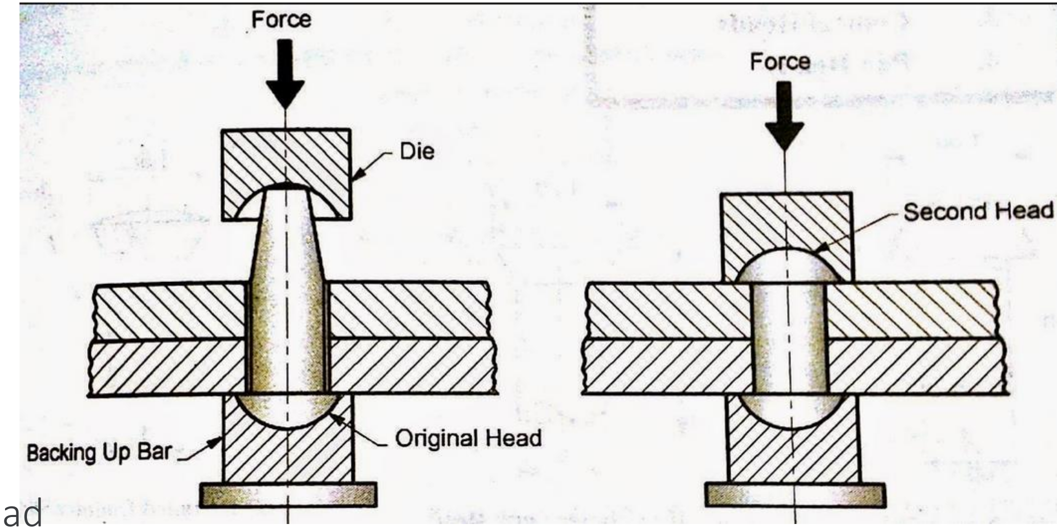
- Pros
  - Low part cost
  - Low profile
  - Wide range of head types
  - Joining dissimilar materials
  - No thermal related issues
  - No damage to surface finish
- Cons
  - Back side access is required
  - Requires both top and bottom side tooling
  - Higher installation cost
  - Moderate resistance to vibration and impact

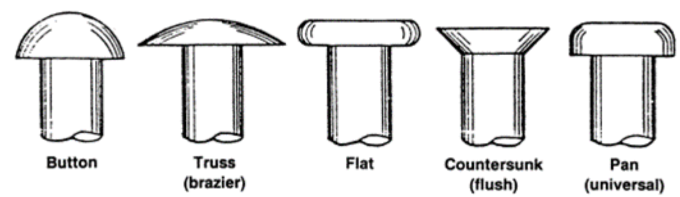




Solid Rivets Pros / Cons

- Pros
  - Low part cost
  - Low profile
  - Wide range of head types
  - Joining dissimilar materials
  - No thermal related issues
  - No damage to surface finish
  - Highest strength due to solid rivet head
- Cons
  - Back side access is required
  - Requires both top and bottom side tooling
  - Higher resistance to vibration and impact
  - Loud impact process for installation





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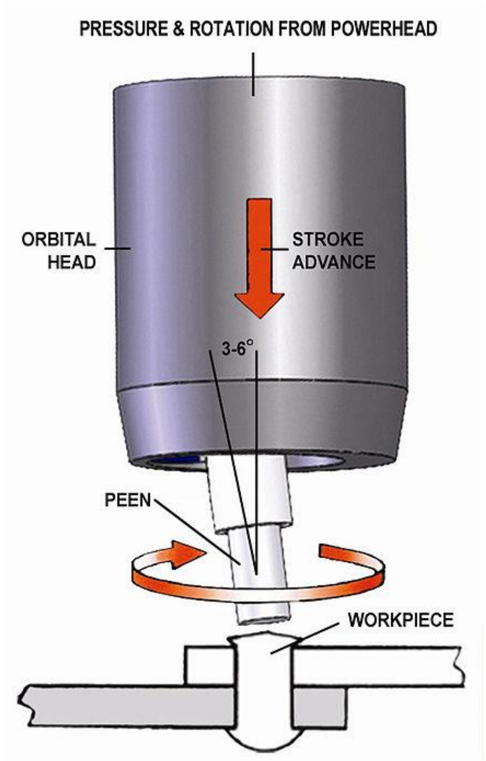
Orbital Riveting Pros / Cons

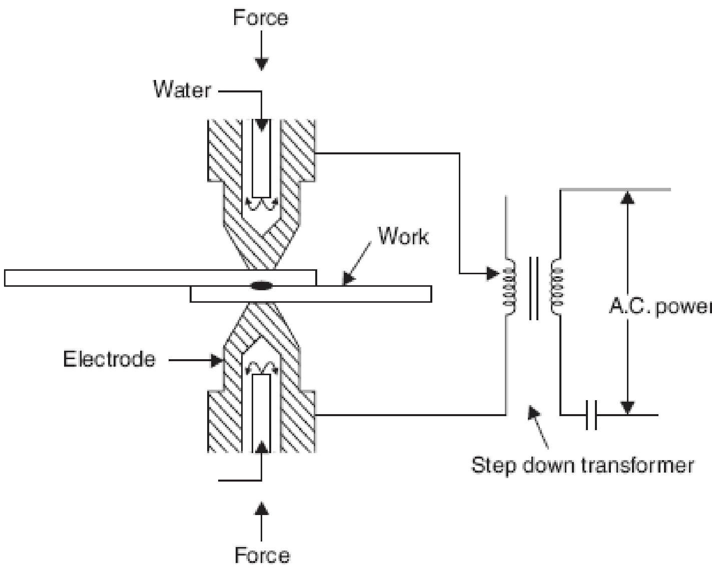
- Pros
  - Tight joints
  - Flexible, can be used to attached shafts for gears, etc.
  - Tightly controlled process
  - No damage to base material
  - Highly resistant to vibration and thermal cycling
- Cons
  - Requires access to both side of the part
  - Space required for spinning head
  - Cycle time (usually between 2 to 15 seconds)

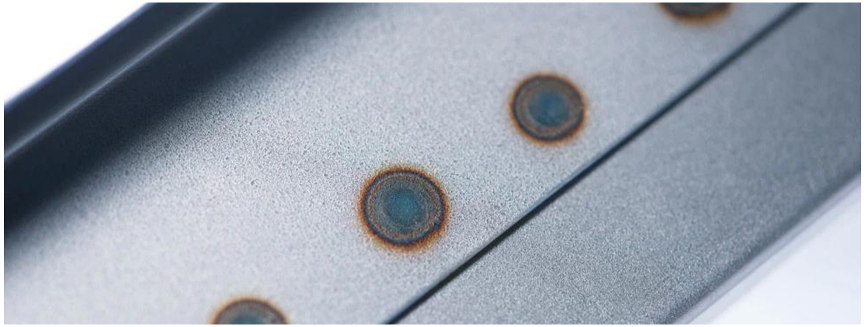
Orbital riveting is a cold-forming process, that relies on high-speed rotation and downward pressure in order to shape the metal.



Spot Welding Pros / Cons

- Pros
  - Capable of welding thin materials
  - Weld dissimilar materials
  - No fillers required
  - Does not require skilled labor
  - Low part distortion
  - High level of welding process control
- Cons
  - Thermal process that generates a heat effected zone
  - Requires a secondary process for corrosion resistance
  - High positional variation w/o tooling

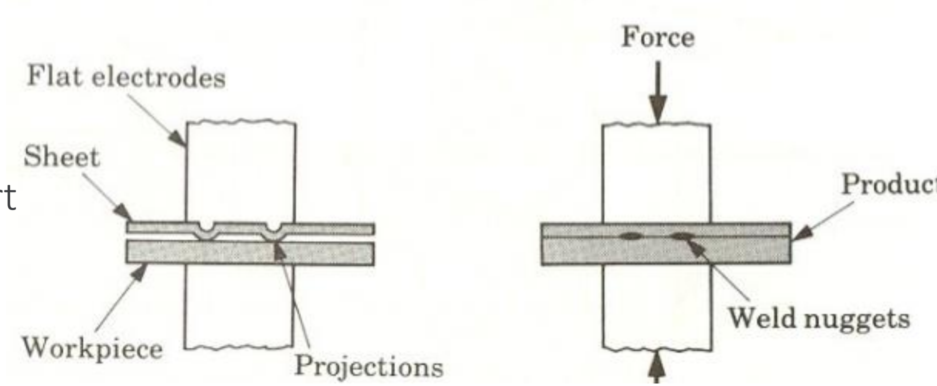




Projection Spot Welding Pros / Cons

- Pros
  - Requires less current than spot welding
  - Efficient – multiple welds in a single operation
  - Not sensitive to material thicknesses
  - No filler materials
  - Better heat balance during welding
- Cons
  - Requires projections to be formed in the part
  - Process required tighter controls

Projection welding a variation of spot welding. Unlike spot welding, electrodes are not used for concentration of heat the projection(s) on the workpiece are used for this purpose.



Before Welding   After Welding

Laser Spot Welding Pros / Cons

- Pros
  - Small weld puddle and short weld time
  - No filler material is required
  - Minimum heat effected zone
  - Easy to automate
  - Can weld complex geometry
- Cons
  - Requires fixtures to achieve required part contact
  - Small weld nugget requires large number of welds
  - Parts need to be designed for this process (tee and lap joints preferred)
  - Requires skilled labor and high investment

Laser welding is a process in which lasers are used to melt the material on the surface of a workpiece, allowing it to be joined to another workpiece made of the same material.

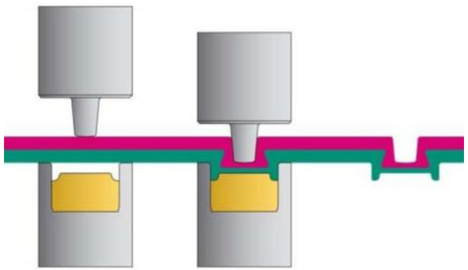
Tox® Pros / Cons

- Pros
  - No heat
  - Connection of 2 or 3 layers of material
  - 30%-60% cost savings vs. spot welding
  - Strain hardening of the material during the process produces a strong joint with high dynamic strength
  - Different sheet thickness and materials in a single joint
  - Integrity of corrosion coating not damaged
  - High level of process control
- Cons
  - High positional variation w/o tooling

TOX® - point diameter

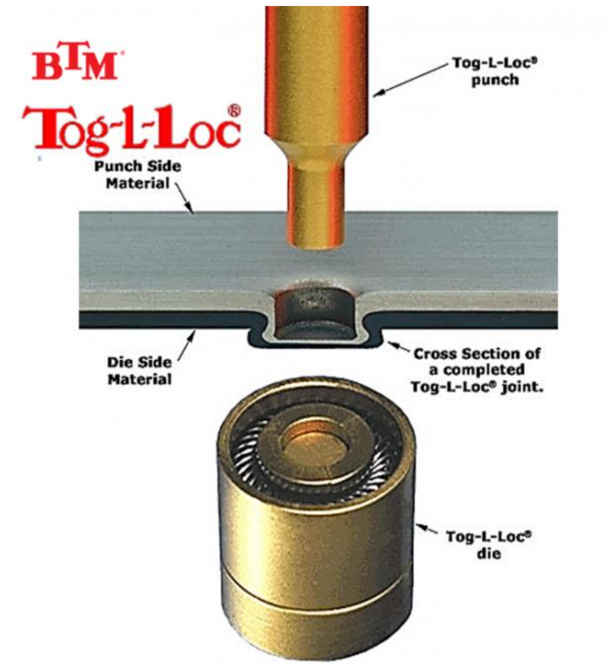
	6 mm	8 mm	10 mm
Single sheet thicknesses Range (mm) Steel	**0.5-1.75**	**1.0-2.5**	**1.25-3.0**
Shear tension (N)	1000-2500	2600-3600	3000-6000
Cross tension (N)	1000-2700	2100-4000	3000-5000
Pressing force (kN)	20-45	35-50	60-80
Stripping force punch side (N)	500 - 3500	1000 - 6000	2000 - 8000

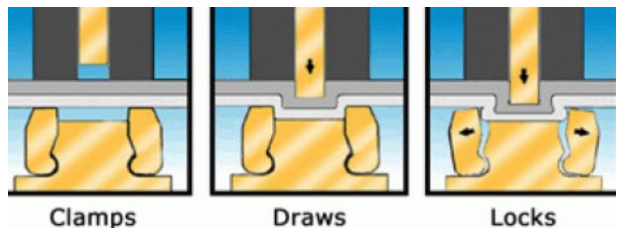
Reference values for TOX®-Joints



Tog-L-Loc® Pros / Cons

- Pros
  - No heat
  - Connection of 2 or 3 layers of material
  - 30%-60% cost savings vs. spot welding
  - Strain hardening of the material during the process produces a strong joint with high dynamic strength
  - Different sheet thickness and materials in a single joint
  - Integrity of corrosion coating not damaged
  - High level of process control
- Cons
  - High positional variation w/o tooling





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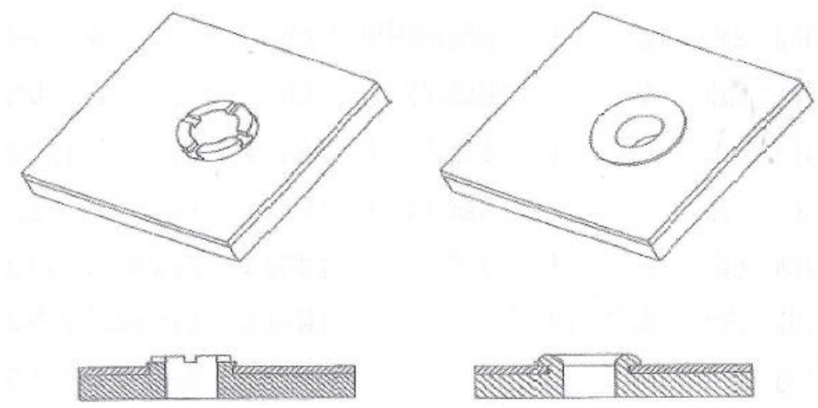
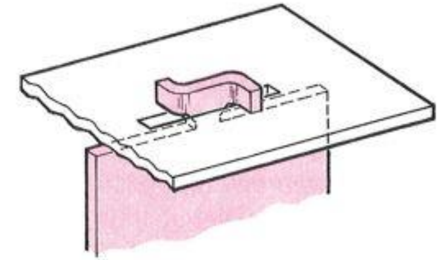
Extrude and Flare Pros / Cons

- Pros
  - No material added
  - Mechanical interlock
  - Good part alignment

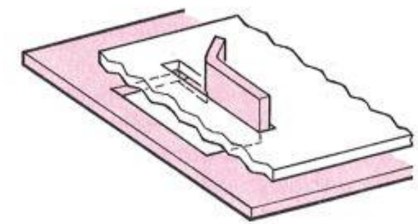
- Cons
  - Limited flare contact depending on flare method



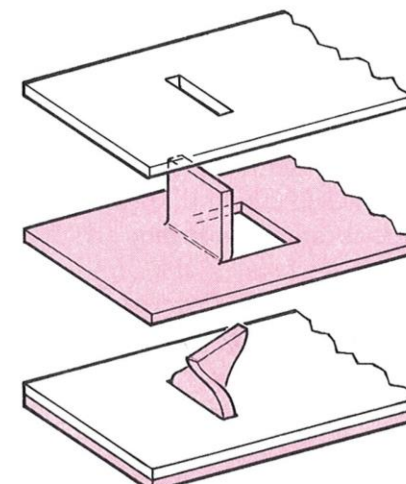
Tab and Fold Joints Pros / Cons



- Pros
  - No additional fasteners
  - Low cost
  - Joins dissimilar thickness and type of materials



- Cons
  - Require tooling
  - Can require a large joint to achieve strength
  - Limited ability to produce a robust tight joint



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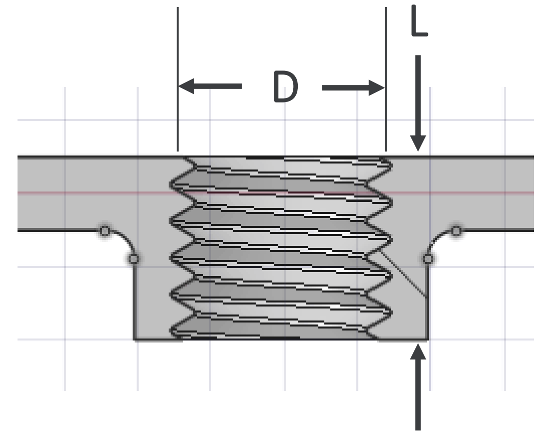
Tab & Screw Joint Pros / Cons

- Pros
  - ~80% of joint strength achieved if extruded barrel length is greater than the screw diameter L > D
  - Joints are temporary
  - Blind joint
  - Wide range of fastener options
  - High reliability
  - Easy to assembly
- Cons
  - Require a threaded hole, insert, or the use of a self thread forming fastener
  - Additional part(s)



CHAPTER 7
MATERIALS

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Material Comparison Normalized

Material	Normalized Cost
Cold Rolled Steel	1
Steel EG	1.2
Aluminum	3
Stainless Steel	6
Copper/Brass	10